Predictive or Also Prognostic? Whether to Include the Expectancy Term in the Hybrid-Design Interaction Model

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Author. pmsimstats team

Purpose. A candidate biomarker such as standing blood pressure may be *predictive*: it changes the effect of prazosin. It may also be *prognostic*: it is associated with the symptom score whatever the treatment, through the natural course, the placebo response or the baseline severity.

This paper asks one practical question for the analysis of a Hybrid N-of-1 trial: should the interaction model include the expectancy term and its interaction with the biomarker, `De + bmc:De`?

It sets out where in the symptom model a biomarker can act, derives what each candidate analysis assumes about it, gives the simulation evidence from the stress test (`docs/46`, `docs/48`), and ends with a decision framework and a recommendation. Results still running are marked as pending.

Update (2026-10-07). The three runs that were pending when this paper was written (baseline-severity cells, F1 with an AIC-selected half-life, F1 with phase-specific variances) are complete. Their results are entered in Sections 3 to 7, with dated notes; the recommendation is unchanged (`docs/46`, Amendment 2).

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Notation index and glossary

Notation follows `analysis/report/NOTATION.md`, `docs/45` and `docs/48`. Symbols marked *local* are defined here.

Table 1: Notation index and glossary of symbols used in this document

Symbol	Meaning	Status
Y_{it}	symptom score (code `Sx`); $t = 0$ is baseline	canonical
BL_i, BR_{it}, PB_{it}, TV_{it}	baseline level; drug, expectancy and natural-course responses	canonical
B_i, b_i	biomarker (code `bm`, centered `bmc`); standardized biomarker	canonical
$D_{bc,it}$	exposure-decayed drug indicator (code `Dbc`)	canonical
η_t	expectancy: 0 at baseline, 1 in the open-label phase, 0.5 when blinded (code `De`)	canonical (NOTATION.md)
β_B, $\beta_{bm:D}$	biomarker main effect; biomarker-by-drug interaction	local (`docs/41`); canonical
$\beta_{bm:e}$	coefficient of `bmc:De`, the biomarker-by-expectancy term	local
$\beta_{B,0}$	coefficient of `bmc:bl0`, the extra biomarker slope at baseline in ES	local
s_0, s_t	true slope of Y on B at baseline, and at post-baseline visit t	local (`docs/48`)
s_{off}, s_{prog}, f	true slope at post-baseline off-drug visits; prognostic slope; share of it absorbed by a shared β_B	local (`docs/48`)
c_{bm}	predictive coupling: biomarker-BR correlation on drug	canonical
g_t	graded coupling weight: 1 on drug, $2^{-t_{sd}/t_{1/2}}$ off drug after exposure, 0 before	local (`docs/37`)
$c_{bm,TV}$, $c_{bm,PB}$, $c_{bm,BL}$	prognostic couplings: biomarker correlation with TV, PB, BL	local
β_{bm}^{TV}	biomarker moderation of the natural-course mean	canonical (component slope)
$G_{TV}(t)$	Gompertz mean curve of the natural course	local
σ_{BR}	SD of the drug response (8 in the stress-test construct)	canonical
σ_{TV}, σ_{BL}	SDs of the natural course (10) and of the baseline level	local
π	power	canonical

Analyses (all with a random intercept, `corCAR1` residuals, CR2 standard errors, linear time, baseline as a response row, and $D_{bc,it}$ at an assumed half-life of 0.5 week):

- **S**: `Sx ~ bmc + t + Dbc + bmc:Dbc`.
- **ES**: S plus `bmc:bl0`, a separate biomarker slope at baseline.
- **F1**: S plus `De + bmc:De`, the expectancy and its interaction with the biomarker.
- **F2**: S plus `bmc:t`, a biomarker slope that changes linearly in time.

Glossary.

- **Predictive biomarker.** Associated with the size of the treatment effect [1].
- **Prognostic biomarker.** Associated with the outcome irrespective of treatment.
- **Constancy assumption.** The biomarker's association with the outcome is the same at baseline as at post-baseline off-drug visits ($s_0 = s_{\text{off}}$).
- **Size.** False-positive rate of the interaction test, nominal 0.05.

1. Summary

1. **Where the biomarker acts decides whether the interaction test is valid.** In the component model a biomarker can act on four components:

   - the drug response (predictive);
   - the natural course or the placebo response (prognostic, post-baseline only);
   - the baseline level (prognostic, at every visit).

2. **The analysis without the expectancy term (S) assumes the biomarker is purely predictive,** or prognostic only through the baseline level. It shares one biomarker slope between baseline and the post-baseline off-drug visits. When the biomarker is prognostic for the natural course or the placebo response, that slope is wrong at one set of visits, and the error is absorbed by the interaction.

3. **Adding `De + bmc:De` (F1) removes the bias in every prognostic scenario simulated.** The expectancy is 0 at baseline, so the term frees the baseline slope. It also lets the slope scale with expectancy, which is exactly the form a placebo-linked biomarker takes. In both scenarios the interaction is then identified from the blinded visits, where on-drug and off-drug visits share the same expectancy.

4. **Evidence** (verified, stress test, 1,000 null replicates per cell):

Table 2: Null rejection rates of S, F1 and ES by biomarker linkage, stress test

Biomarker linked to	S	F1	ES
natural course, constant	0.099	0.047	0.049
natural course, growing	0.049	0.046	0.063
placebo response	0.100	0.040	0.049
baseline severity (added 2026-10-07)	0.050	0.040	0.050
nothing (11 standard cells, pooled)	0.048	0.051	0.048

5. **The cost is power, and it depends on the correlation structure.** Under AR(1) data F1 gives up about 30% of S's power (0.364 against 0.514 at $c_{bm} = 0.30$, size-adjusted). Under compound symmetry the loss is small (0.522 against 0.554 at $c_{bm} = 0.25$).

6. **The direct evidence on blood pressure is weak.** In the placebo arm of the trial that motivated blood pressure as a biomarker, baseline pressure showed no association with symptom change [2]. That arm had 35 participants, so the evidence cannot rule out a prognostic association of the size that biases S.

7. **Recommendation.** Include the expectancy term in the primary analysis (F1), with S and ES as sensitivity analyses. A material disagreement between F1 and S is itself evidence that the biomarker is prognostic. Reverse the roles only if independent data show that the biomarker is not prognostic.

2. Where a biomarker can act

The symptom score is $Y_{it} = \mathrm{BL}_i - (TV_{it} + PB_{it} + BR_{it})$. Each component can carry an association with the biomarker. The table gives the implied slope of Y on b_i at baseline and at post-baseline visits.

Table 3: Biomarker pathways by kind with implied baseline and post-baseline slopes

Pathway	Kind	s_0 (baseline)	s_t, post-baseline	Shape over visits
Drug response, c_{bm}	predictive	0	$-c_{bm}\sigma_{BR}\, g_t$	full on drug, decaying off drug

Pathway	Kind	s_0 (baseline)	s_t, post-baseline	Shape over visits
Natural course, $c_{bm,TV}$	prognostic	0	$-c_{bm,TV}\sigma_{TV}$	constant after baseline
Natural course, β_{bm}^{TV}	prognostic	0	$-\beta_{bm}^{TV}\sigma_{TV}\,G_{TV}(t)$	grows with the natural course
Placebo response, $c_{bm,PB}$	prognostic	0	$-c_{bm,PB}\cdot 10\,\eta_t$	scales with expectancy
Baseline severity, $c_{bm,BL}$	prognostic	$c_{bm,BL}\sigma_{BL}$	$c_{bm,BL}\sigma_{BL}$	constant at every visit, baseline included

Slopes per SD of biomarker, derived from the stress-test construct (configuration C at $\rho = 0.7$; `docs/46`, Section 5): $\sigma_{TV} = 10$, the PB SD is $10\eta_t$, and g_t is the graded coupling.

Only the first row is the estimand. In every other row the biomarker's association does not depend on drug, so a correct analysis must report no interaction. Two features of the Hybrid design make some of these pathways dangerous.

- **Baseline precedes every response component.** A prognostic association through the natural course or the placebo response is zero at baseline and nonzero afterwards. A baseline-severity association is the same at every visit.
- **Expectancy and drug are confounded in the open-label phase.** Every open-label visit is on drug and has $\eta_t = 1$, while every off-drug visit is blinded with $\eta_t = 0.5$. A slope that scales with expectancy is therefore larger on drug than off drug on average, even with no drug interaction.

3. What each analysis assumes

Write the biomarker slope the analysis fits at each kind of visit. With $D_{bc} = 1$ on drug and close to 0 off drug:

Table 4: Biomarker slope fitted by S, ES and F1 at each kind of visit

Visit	S	ES	F1
baseline	β_B	$\beta_B + \beta_{B,0}$	β_B

Visit	S	ES	F1
open label, on drug	$\beta_B + \beta_{bm:D}$	$\beta_B + \beta_{bm:D}$	$\beta_B + \beta_{bm:e} + \beta_{bm:D}$
blinded, on drug	$\beta_B + \beta_{bm:D}$	$\beta_B + \beta_{bm:D}$	$\beta_B + 0.5\beta_{bm:e} + \beta_{bm:D}$
blinded, off drug	β_B	β_B	$\beta_B + 0.5\beta_{bm:e}$

Here $\beta_{B,0}$ is the coefficient of `bmc:bl0`.

S ties baseline to the off-drug visits. Under any pathway with $s_0 \neq s_{\text{off}}$, the fitted β_B falls between the two true slopes. The on-drug rows then differ from it by part of the prognostic slope, which is attributed to $\beta_{bm:D}$ (`docs/48`, Section 2): $\hat{\beta}_{bm:D} \approx (1 - f)\, s_{\text{prog}}$, with f set by the weight of the baseline row.

ES frees baseline but assumes one post-baseline off-drug slope. It is exact when the prognostic slope is constant after baseline.

F1 frees baseline through $\eta_0 = 0$, and lets the slope change with expectancy. Under an expectancy-scaled pathway ($s_t \propto \eta_t$), the `bmc:De` term absorbs it exactly. Under a constant post-baseline pathway, F1 misfits the open-label and blinded slopes. But the misfit is the same at blinded on-drug and blinded off-drug visits, so it cancels in the blinded contrast that then identifies $\beta_{bm:D}$ (derived from the table). The cancellation is exact for a weighted least-squares fit of the four slopes, with visits treated as independent and $D_{bc} = 0$ off drug, when the blinded on-drug and off-drug visits carry equal total weight: the normal equations then give $\hat{\beta}_{bm:D} = 0$ whatever the weights of the baseline and open-label visits. With unequal weights, or under serial correlation, a residual bias can remain (derived); the simulated rejection rates of 0.046 and 0.047 suggest it is small in this design.

(Note 2026-10-08.) The equal-weight condition does not hold in the Hybrid design: the blinded on-drug and off-drug visits carry total weights 176 and 244, so F1 is only approximately unbiased under a constant post-baseline pathway. The limit is $\hat{\beta}_{bm:D} \to s_{\text{prog}}(1/w_{\text{off}} - 1/w_{\text{on}})/\sum_k 1/w_k$ (derived). Per unit of s_{prog} the residual bias is -0.051 with binary D and independent visits, -0.020 with D_{bc}, 0.089 under a random-intercept CAR(1) working covariance and 0.171 under the true AR(1) covariance (verified, computed in the published construct of `04-power-simulation.R`, not the stress-test construct; `analysis/report/compendium-mathematics.tex`, Section on the prognostic biomarker). The simulated 0.047 is consistent with a bias of this size. In both cases the interaction rests on the comparison the Hybrid design was built to make clean: blinded on drug against blinded off drug, at the same expectancy.

Baseline severity gives $s_0 = s_t$ at every visit, so the constancy assumption holds and S should be unbiased. This is a prediction; the simulation is running (Section 5). (Confirmed 2026-10-07: S rejects 0.050 in the baseline-severity null cell; Section 5.)

4. What F1 gives up

F1 estimates two biomarker slopes that S constrains. Its interaction then draws on less of the data:

- the open-label visits mostly inform $\beta_{bm:e}$ rather than $\beta_{bm:D}$;
- the baseline row no longer informs the off-drug slope.

Under AR(1) data the baseline row is a valuable observation of the off-drug slope (`docs/45`, Section 6). Losing it costs about 30% of the power. Under compound symmetry the participant effect already supplies that information, and the loss is small.

Size-adjusted power (verified, stress test, 500 replicates per cell):

Table 5: Size-adjusted power of S, F1 and ES by stress-test cell

Cell	S	F1	ES
AR(1), $c_{bm} = 0.30$	0.514	0.364	0.364
AR(1), $c_{bm} = 0.45$	0.808	0.672	0.656
Compound symmetry, $c_{bm} = 0.25$	0.554	0.522	0.662
Natural course prognostic, constant, $c_{bm} = 0.30$	(biased)	0.482	0.370
Natural course prognostic, growing, $c_{bm} = 0.30$	(nominal) 0.516	0.378	0.192
Placebo prognostic, $c_{bm} = 0.30$	(biased)	0.446	0.498
Baseline-severity prognostic, $c_{bm} = 0.30$ (added 2026-10-07)	(nominal) 0.444	0.396	0.336

Two observations:

- **ES and F1 are close when the biomarker is purely predictive.** F1 keeps its power under every prognostic form, whereas ES loses most of it when the natural-course association grows over time.
- **S's power in the prognostic cells is not a fair comparison.** Its test is biased there; "size-adjusted" corrects the level, not the bias.

5. Pending results (completed 2026-10-07)

Three post hoc runs on the same simulated trials are in progress (`11-strawman-stress-test.R`):

1. **Baseline severity** ($c_{bm,BL} = 0.3$, null and alternative) under every analysis. Prediction: all analyses nominal, because the constancy assumption holds.
2. **F1 with an AIC-selected half-life** (F1A). It tests whether the robust model keeps its size, and how much power data-driven selection costs it (about 0.03 for S).
3. **F1's terms with phase-specific residual variances** (G1). It tests whether part of the power F1 gives up can be recovered while keeping its robustness.

These results will be added here when they are in. They are post hoc: they can motivate a change in recommendation but not settle it.

Results (2026-10-07; verified, `decision-*-ds-nosym-all14.csv`, `docs/46`, Amendment 2).

Table 6: Size and size-adjusted power of F1 and the three post hoc analyses

Analysis	Pooled null, 11 standard cells	Worst standard cell	Prognostic null cells (4)	AR(1), $c_{bm} = 0.30$	CS, $c_{bm} = 0.25$
F1	0.051	0.062	0.040 to 0.047	0.364	0.522
F1A (F1, AIC half-life)	0.057	0.073 ($t_{1/2} = 1$)	0.045 to 0.059	0.338	0.446
G1 (F1, phase variances, CR2)	0.049	0.060	0.042 to 0.055	0.354	0.546

Analysis	Pooled null, 11 standard cells	Worst standard cell	Prognostic null cells (4)	AR(1), $c_{bm} = 0.30$	CS, $c_{bm} = 0.25$
G2 (F1, phase variances, model-based)	0.047	0.066	0.043 to 0.062	0.380	0.528

1. **Baseline severity.** The prediction holds. In the null cell every analysis is within the cell limit of 0.075, from 0.040 (F1) to 0.075 (the published analysis); S rejects 0.050. In the alternative cell S has the higher power (0.444 against 0.396 for F1), as expected when the constancy assumption holds.
2. **F1A fails the size criterion.** Its pooled rate of 0.057 is outside the 2-MCSE band (0.046 to 0.054), and it rejects 0.073 when $t_{1/2} = 1$. With the expectancy terms in the model, selecting the half-life by AIC is anti-conservative, and it also costs power (0.026 at $c_{bm} = 0.30$ in the AR(1) construct). It is not recommended.
3. **G1 and G2 keep F1's robustness but add little power.** Both are nominal in every standard and prognostic null cell. Averaged over the 16 non-prognostic alternative cells of the AR(1) construct they gain 0.002 (G1) and 0.017 (G2) over F1. G2 and F1 tie within 0.02 on maximin shortfall (0.232 against 0.242), and the tie goes to F1, which has fewer covariance parameters (`docs/48`, Section 5).

6. Decision framework

Question 1: is there independent evidence that the biomarker is not prognostic? For blood pressure, the only direct evidence is a null placebo-arm association in 35 participants [2]. That cannot exclude a prognostic slope large enough to bias S. A prognostic coupling of 0.3 roughly doubles S's false-positive rate. Without stronger evidence, treat the biomarker as possibly prognostic.

Question 2: which pathway is plausible?

- **Natural course or placebo response.** A noradrenergic marker could plausibly track spontaneous improvement or placebo response. F1 held its size under all three forms simulated (0.040 to 0.047); under the constant form this is approximate, not exact (Section 3).
- **Baseline severity only.** S is valid (0.050 in the simulated null cell, Section 5), but this does not remove the risk from the other pathways.

Question 3: what is the cost of protection? It depends on the serial correlation the trial is likely to show. If pilot data suggest little serial correlation beyond a participant effect, F1 costs little. Under strong serial correlation it costs about 30% of the power. The sample size should then be planned for F1, not for S.

Question 4: how should disagreement be read? Report S as a sensitivity analysis. If F1 and S agree, the conclusion does not depend on the constancy assumption. If they disagree materially, the disagreement is itself evidence of a prognostic association. Report F1, and report the estimate of `bmc:De` as an exploratory measure of it.

7. Recommendation

1. **Primary analysis: F1.** Random intercept with `corCAR1` residuals, CR2 standard errors, linear time, baseline as a response row, D_{bc} at a fixed half-life of 0.5 week, and `De + bmc:De`.
2. **Sensitivity analyses:**
   - S (constancy assumed);
   - ES (baseline freed, expectancy not modeled);
   - binary D in place of D_{bc}.
3. **Plan the sample size for F1,** with the serial-correlation assumption stated.
4. **Revisit when the pending runs are in.** F1 with an AIC-selected half-life may replace the fixed half-life on grounds of face validity, at a power cost to be measured. The phase-variance version of F1 may replace F1 if it holds size everywhere with materially more power. Either change would need an independent run. (Revisited 2026-10-07, Section 5: F1 with AIC selection fails the size criterion, and the phase-variance versions hold size but add no material power. F1 with the fixed half-life of 0.5 week remains the primary analysis.)

8. Limitations

- **One strength per prognostic form.** All prognostic forms were simulated at 0.3; how bias grows with the strength is not mapped.
- **Single pathways only.** Combinations, for example natural course and placebo together, were not simulated.
- **Linear, Gaussian couplings.** A biomarker that acts on the rate rather than the size of response is outside the stress-test construct.
- **One design.** The expectancy argument is specific to the Hybrid design's confounding of open-label and on-drug visits. In a fully blinded crossover the expectancy term would be constant and unnecessary.

- **The mechanism is heuristic.** The account of Section 3 explains the observed pattern but is not a derivation of the bias for each analysis.

9. Reproducibility

```
Rscript analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R \
  --cores 8 --chunk 25
Rscript analysis/scripts/quick-sim/carryover-closed-form/12-strawman-stress-decision.R \
  --strawman DS --no-symmetry
# post hoc analyses (G1, G2; F1A) and the 14-analysis decision
Rscript analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R \
  --cores 8 --chunk 25 --extra
Rscript analysis/scripts/quick-sim/carryover-closed-form/11-strawman-stress-test.R \
  --cores 8 --chunk 25 --extra2
Rscript analysis/scripts/quick-sim/carryover-closed-form/12-strawman-stress-decision.R \
  --strawman DS --no-symmetry --with-extras
```

Outputs are in `analysis/data/quick-sim/carryover-closed-form/strawman-stress/`.

10. References

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3. Liu GF, Lu K, Mogg R, Mallick M, Mehrotra DV. Should baseline be a covariate or dependent variable in analyses of change from baseline in clinical trials? *Statistics in Medicine* 2009; 28(20):2509-2530. doi:10.1002/sim.3639.
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